The Leather Jacket Stays On: Jensen's Inequality for Convex Combinations of Huangs

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Abstract

Jensen's inequality states that for a convex function φ and an integrable random variable X, $\varphi(\mathbb{E}[X]) \le \mathbb{E}[\varphi(X)]$, with equality, for strictly convex φ, if and only if X is almost surely constant. On November 8, 2026, Jensen confirmed that he is not. We develop the resulting *convex analysis of Huangs*. We prove that the set of Huangs is HUANG (**H**ull **U**nchanged by **A**veraging, **N**othing **G**ained): it is closed under convex combination, so that every λ-mixture of LORI (**L**ocal **O**ptimum, **R**eally the **I**nfimum) and ALICE (**A**llegedly **L**atent, **I**nterior, **C**ompromising **E**ndpoint) is again a Huang, and the mean Huang is a Huang. We introduce the *Leather Jacket* $\mathscr{J}(S) = \mathrm{conv}(S)$ and prove that it is a closure operator—extensive, monotone, and idempotent—so that the leather jacket stays on. For two-point Huang measures we characterise the *Huang gap* $\Delta_\varphi(\lambda) = \mathbb{E}[\varphi(H)] - \varphi(\mathbb{E}[H])$: it vanishes exactly at the vertices (monogamy), is concave in λ, and for $\varphi = \|\cdot\|^2$ equals $\lambda(1-\lambda)\|\ell - \alpha\|^2$, the *variance of Huangs*, maximised under the principle of insufficient reason at $\lambda = \frac{1}{2}$. We prove that strict convexity is unidentifiable from two photographs and identifiable from three, bound the dimension of Jensen by the number of Huangs minus one via Carathéodory's theorem, and show that a love triangle is a 2-simplex. Finally we perform convexity analysis with Chinese characteristics, modelling the defence that Jensen "meets thousands of customers" as a Chinese Restaurant Process whose expected number of tables grows as $\alpha \log n$. All bounds are tight. All proofs are trivial by Jensen's.

*No relation.

†Also no relation. By Theorem 1, any convex combination of the authors is a third author, who has asked not to be listed.

1 Introduction: Taipei 101

Sur les fonctions convexes et les inégalités entre les valeurs moyennes.
—J. L. W. V. Jensen, 1906 [1]

I was unfaithful to my wife.
—J. H. Huang, 2026 [15]

In 1906 the Danish engineer Johan Jensen, employed by the Copenhagen Telephone Company and doing mathematics in his spare time, proved that for every convex function φ and every integrable random variable X,

$$\varphi\big(\mathbb{E}[X]\big) \;\le\; \mathbb{E}\big[\varphi(X)\big]. \tag{1}$$

The inequality is named after him. One hundred and twenty years later a different engineer named Jensen,[1] employed by NVIDIA and doing other things in his spare time, provided the first large-scale empirical confirmation of its *equality case*. Recall that for strictly convex φ, equality holds in (1) if and only if X is almost surely constant; more generally, equality holds if and only if φ is affine on the support of X [3, 4]. Before 2026 this was known as the trivial case. We now call it the *faithful* case.

The present paper asks the natural question: *when X is a Huang, how strict is Jensen's inequality?* Our answer, developed over the following sections, is: strictly. Specifically, we make the following contributions.

- We introduce the Leather Jacket $\mathscr{J}(S) = \mathrm{conv}(S)$ and prove that it is a closure operator (Theorem 11): the leather jacket stays on.
- We prove that the set of Huangs is convex (Theorem 1): any convex combination of Huangs is a Huang. The mean Huang is a Huang.
- We prove Jensen's inequality in *personal form* (Theorem 3) and deduce from the memo of November 8,

[1] Jen-Hsun on legal documents [22]; Jensen on everything else, including, we assume, this one.

2026 [15] that it is strict (Corollary 4): Jensen has, in his own words, "fallen short."

- We characterise the two-point *Huang gap* (Theorem 5). It vanishes exactly at the vertices, which we call monogamy, and for quadratic scrutiny it equals the *variance of Huangs*, which we compute to be a quarter of the Pacific Ocean squared.
- We prove that strict convexity cannot be detected from two photographs (Taipei, 2024 and 2025) and can be detected from three (Theorem 6). The third photograph was supplied by the New York Times [17].
- We show that responsibility is a vertex while practice is an interior point (Theorem 9), and that a love triangle is a 2-simplex (Definition 7).
- We perform convexity analysis with Chinese characteristics (Section 5), including the first application of the Chinese Restaurant Process to the NVIDIA investor-relations department.
- We provide a comparison table with a Trivago row.

Why Taipei 101. Taipei 101 has 101 floors. This section is the ground floor. The photographs that began this line of research were taken in Taipei in 2024 and 2025 [21]; the convex combinations they document are analysed on the floors above.

2 Related Work

2.1 Two engineers named Jensen

Johan Jensen (1859–1925) was an engineer at the Copenhagen Telephone Company who did mathematics as a hobby and proved an inequality [1]. Jen-Hsun Huang (1963–) is an engineer at NVIDIA who was *proven to be* one [15]. Both Jensens worked in telecommunications in the broad sense; only one of them communicated with Taipei.

2.2 Huang's Law versus Jensen's inequality

Huang's Law [11] is the observation that GPU performance more than doubles every two years, outpacing Moore's Law. It is stated as an equality. Jensen's inequality is, as its name suggests, an inequality. We show in this paper that the man himself is strict (Corollary 4). The only inequality Jensen has consistently respected is "the more you buy, the more you save," which we prove is Jensen's inequality for concave cost (Claim 13).

2.3 The Jensen gap

The quantity $\mathbb{E}[\varphi(X)] - \varphi(\mathbb{E}[X])$ is known as the *Jensen gap*, and its second-order behaviour $\tfrac{1}{2}\varphi''(\mu)\sigma^2$ is known as *curvature bias*. The authors have prior work on this gap in an optically thin plasma [14], which the reviewers of the present paper would not enjoy. The authors' earlier SIGBOVIK work established that classification accuracy is "trivial by Jensen's" [12]; we extend this proof technique to every theorem in this paper. Separately, the authors established that the most important thing in life will always be the people in this room [13]. Jensen was in a different room.

2.4 Honeypots

In computer security a *honeypot* is a decoy deployed by the defender to attract an attacker. In the matter at hand the roles are, allegedly, reversed. We note for completeness that the ALICE of this paper is not to be confused with the Alice of cryptographic protocols, who is honest by convention. Allegedly, this Alice was Eve.

3 Preliminaries

Throughout, $\mathcal{H} \subset \mathbb{R}^d$ denotes the set of Huangs. We distinguish three elements: $\ell \in \mathcal{H}$ (Lori), $\alpha \in \mathcal{H}$ (Alice), and $j \in \mathcal{H}$ (Jensen). The scalar $\lambda \in [0,1]$ is the weight placed on Lori. A *Huang* H is a $\mathcal{H}$-valued random variable. *Scrutiny* is a strictly convex function $\varphi : \mathbb{R}^d \to \mathbb{R}$; we assume strict convexity because scrutiny increases faster than linearly in the distance from Lori, an assumption validated in November 2026.

Definition 1 (LORI)**.** ℓ is a **L**ocal **O**ptimum, **R**eally the **I**nfimum.

Remark 1. For a convex objective, every local optimum is global [3]. Lori has been a local optimum since 1985, having previously been Jensen's engineering lab partner at Oregon State [22]; by convexity she is therefore the infimum, and the infimum is attained because a marriage is closed and bounded (41 years) and hence compact.

Definition 2 (ALICE)**.** α is an **A**llegedly **L**atent, **I**nterior, **C**ompromising **E**ndpoint.

Remark 2. *Latent:* her whereabouts are reported as unknown [21]; we therefore marginalise over them. *Interior:* by Slater's condition, the existence of a strictly interior feasible point guarantees strong duality [4], i.e. that the dual (the Board) attains the same value as the primal (Jensen).[2] *Compromising:* the Board has "no reason to believe" that anything was [16]; we adopt this as our prior. *Endpoint:* of the segment $[\ell, \alpha]$. The word *allegedly* distributes over the remaining four.

Definition 3 (HUANG property)**.** A set $S \subseteq \mathbb{R}^d$ is HUANG (**H**ull **U**nchanged by **A**veraging, **N**othing **G**ained) if $\operatorname{conv}(S) = S$, i.e. if averaging its elements with nonnegative weights summing to one gains nothing new.

[2] Megaspeed, her reported employer [19], is also the regime of step sizes $\eta > 2/L$ at which gradient descent diverges on even a convex quadratic.

A set is HUANG if and only if it is convex. We use the two words interchangeably and, as it turns out, we are entitled to.

Definition 4 (The JENSEN operator)**.** For $\lambda \in [0,1]$ and $x, y \in \mathbb{R}^d$, the JENSEN operator (**J**acket-**E**nclosed **N**onnegative **S**um, **E**xactly **N**ormalised) is $\textsc{Jensen}_\lambda(x,y) = \lambda x + (1-\lambda)y$. More generally, $\sum_i \lambda_i x_i$ with $\lambda_i \geq 0$ and $\sum_i \lambda_i = 1$ is a *convex combination*.

Definition 5 (The Leather Jacket)**.** The Leather Jacket of $S \subseteq \mathbb{R}^d$ is its convex hull,

$$\mathscr{J}(S) = \operatorname{conv}(S) = \left\{ \textstyle\sum_{i=1}^m \lambda_i x_i : \; x_i \in S,\; \lambda_i \geq 0,\; \sum_i \lambda_i = 1 \right\}$$

(finite sums), equivalently the intersection of all convex sets containing S: the smallest convex set that covers every Huang.

Definition 6 (Huang measure and Huang gap)**.** The two-point *Huang measure* with parameter λ is $\mu_\lambda = \lambda\,\delta_\ell + (1-\lambda)\,\delta_\alpha$. If $H \sim \mu_\lambda$ then $\mathbb{E}[H] = \textsc{Jensen}_\lambda(\ell, \alpha)$. The *Huang gap* of an integrable Huang H under scrutiny φ is

$$\Delta_\varphi(H) \;=\; \mathbb{E}[\varphi(H)] - \varphi(\mathbb{E}[H]).$$

We will test the following hypotheses.

Hypothesis 1 (Affine Jensen)**.** φ is affine on $\operatorname{supp}(H)$.

Hypothesis 2 (Constant Jensen)**.** H is almost surely constant.

Hypothesis 3 (Constant Jacket)**.** The jacket K worn by Jensen on a uniformly random day is almost surely constant.

Hypothesis 4 (Thousands of customers)**.** Jensen meets $n \geq 10^3$ customers [19].

4 Main Results

4.1 Huangs are HUANG

Theorem 1 (Huangs are HUANG)**.** *The set of Huangs $\mathscr{H}$ is convex. Consequently, for every $\lambda \in [0,1]$, $\textsc{Jensen}_\lambda(\ell,\alpha) \in \mathscr{H}$, and for every integrable Huang H, $\mathbb{E}[H] \in \mathscr{H}$.*

Proof. Let $x, y \in \mathscr{H}$ and $\lambda \in [0,1]$. Then $\lambda x + (1-\lambda)y$ is a convex combination of Huangs and is therefore, by inspection of the surname, a Huang. The statement about $\mathbb{E}[H]$ follows because the expectation of a random variable supported on a convex set lies in that set (take the closure if necessary). □

Corollary 2 (The jacket fits)**.** *$\mathscr{J}(\mathscr{H}) = \mathscr{H}$. In particular the mean Huang $\mathbb{E}[H]$ is a Huang. Whether the Huang is mean is outside the scope of convex analysis.*

Remark 3. Three Huangs appear in this paper (Lori, Alice, Jensen). By Theorem 1, any convex combination of them is a fourth. This is the mechanism by which the family name is preserved under the operations described in [15]: Jensen stayed within the family, in the sense of closure under convex combination and in no other sense.

4.2 Jensen's inequality, personal form

Theorem 3 (Jensen's inequality, personal form)**.** *Let φ be strictly convex and let H be an integrable Huang with $\mathbb{E}[|\varphi(H)|] < \infty$. Then*

$$\varphi(\mathbb{E}[H]) \;\leq\; \mathbb{E}[\varphi(H)],$$

with equality if and only if H is almost surely constant.

Proof. Trivial by Jensen's [12]. Readers who require more than this are referred to Appendix A. □

Corollary 4 (Jensen has fallen short)**.** *Hypothesis 2 is rejected: by [15], H is not almost surely constant. Hence Jensen's inequality is strict, $\varphi(\mathbb{E}[H]) < \mathbb{E}[\varphi(H)]$: the value at Jensen's expectation* falls short *of his expected value. This is the precise content of the statement that Jensen has "fallen short" [15], and the right-hand side is what the Board, partners and shareholders "deserve" [15].*

Remark 4 (The one case of equality). Equality in Theorem 3 holds for almost surely constant random variables. Hypothesis 3 is not rejected by any evidence known to the authors: the jacket is black and the jacket is leather. Hence $\varphi(\mathbb{E}[K]) = \mathbb{E}[\varphi(K)]$. The jacket is the unique component of Jensen for which Jensen's inequality is an equality. Moreover, since K is constant, the σ-algebra it generates is trivial, $\sigma(K) = \{\emptyset, \Omega\}$, so $\mathbb{E}[X \mid \sigma(K)] = \mathbb{E}[X]$ for every X: *conditioning on the jacket reveals nothing*. This is consistent with the findings of the Board [16].[3]

4.3 The two-point Huang gap

Theorem 5 (The Huang gap)**.** *Let $H \sim \mu_\lambda$ with $\ell \neq \alpha$ and let φ be strictly convex. Write*

$$\Delta_\varphi(\lambda) = \lambda\,\varphi(\ell) + (1-\lambda)\varphi(\alpha) - \varphi\big(\lambda\ell + (1-\lambda)\alpha\big).$$

Then:

(i) $\Delta_\varphi(0) = \Delta_\varphi(1) = 0$*: the gap vanishes at the vertices (monogamy, with either Huang).*

(ii) Δ_φ *is concave on* $[0,1]$ *and* $\Delta_\varphi(\lambda) > 0$ *for* $\lambda \in (0,1)$*: every interior point has a positive gap, and the gap is maximised at some* $\lambda^* \in (0,1)$*.*

(iii) (Bregman form) $\Delta_\varphi(\lambda) = \mathbb{E}\big[D_\varphi(H, \mathbb{E}[H])\big]$*, where* $D_\varphi(x,y) = \varphi(x) - \varphi(y) - \langle \nabla\varphi(y), x - y\rangle$ *is the Bregman divergence: the Huang gap is the expected divergence of each Huang from the mean Huang.*

(iv) (Variance of Huangs) For $\varphi = \|\cdot\|^2$*,* $\Delta_\varphi(\lambda) = \lambda(1-\lambda)\,\|\ell - \alpha\|^2 = \operatorname{Var}(H)$*, maximised at* $\lambda^* = \frac{1}{2}$ *with value* $\frac{1}{4}\|\ell - \alpha\|^2$*.*

(v) (Curvature bias) If $d = 1$ *and* $m \leq \varphi'' \leq M$ *on* $[\alpha, \ell]$*, then* $\frac{m}{2}\operatorname{Var}(H) \leq \Delta_\varphi(\lambda) \leq \frac{M}{2}\operatorname{Var}(H)$*.*

[3] Jensen does not wear a watch, because "now is the most important time" [22]. He does wear the jacket, because a constant function is invariant under time translation.

Proof. See Appendix A. Each item is trivial by Jensen's in a different way. □

Remark 5 (A quarter of the Pacific squared). Taking $\|\ell - \alpha\|$ to be the great-circle distance from Santa Clara, CA to Taipei, 10,418 km, item (iv) gives a maximal variance of Huangs of $\frac{1}{4}(10{,}418\text{ km})^2 \approx 2.7 \times 10^7\text{ km}^2$. The variance of Huangs grows with the *square* of the distance between them, which is why transpacific Huangs are so much more variable than, say, Huangs in the same engineering lab.

Remark 6 (Principle of insufficient reason). Absent any information about λ, Laplace's principle of insufficient reason [10] prescribes $\lambda = \frac{1}{2}$, which by item (iv) also maximises the variance of Huangs. We note that the affair itself appears to have been conducted under the same principle.

Remark 7 (Entropy). For the negative-entropy function $\varphi(h) = h\log h$ on $\{0,1\}$ (Alice at 0, Lori at 1) one computes $\Delta_\varphi(\lambda) = -\lambda\log\lambda$, the Lori term of the binary entropy, maximised at $\lambda^* = 1/e \approx 0.368$. The number $1/e$ also governs the secretary problem [9], in which one rejects the first 37% of candidates and then commits to the next one who is the best so far. Lori was that candidate in 1985. The algorithm terminates there.

4.4 Curvature is undetectable from two photographs

Theorem 6 (Three photographs)**.** *Let $d = 1$ and suppose the scrutiny φ is observed only through* noiseless photographs *$(h_i, \varphi(h_i))$, $i = 1,\dots,n$.*

(a) If $n = 2$, there exists an affine $\tilde\varphi$ consistent with the photographs. Under $\tilde\varphi$, $\Delta_{\tilde\varphi} \equiv 0$ for every Huang measure supported on $\{h_1, h_2\}$. Hence Hypothesis 1 cannot be rejected from two photographs, whatever they show.

(b) If $n = 3$ with $h_1 < h_3 < h_2$ and the third photograph lies strictly below the chord through the first two, then φ is not affine on $[h_1, h_2]$ and the Huang measure on $\{h_1, h_2\}$ with mean h_3 has $\Delta_\varphi > 0$: Hypothesis 1 is rejected at every significance level.

Proof. (a) Two points determine a line. (b) The point h_3 is the convex combination $\lambda h_1 + (1-\lambda)h_2$ with $\lambda = (h_2 - h_3)/(h_2 - h_1) \in (0,1)$, and the chord value at h_3 is $\lambda\varphi(h_1) + (1-\lambda)\varphi(h_2)$; the difference is exactly $\Delta_\varphi(\lambda)$, which is positive by assumption. Since the photographs are noiseless, the test statistic is zero under the null and positive under the alternative; the p-value is 0. □

Remark 8. Two photographs of Jensen and Alice in Taipei (2024, 2025) were in circulation before October 2026 [21]; by (a), they were consistent with an affine Jensen, and NVIDIA correctly described them as "not unusual" [18]. The third data point was contributed by the New York Times on October 29, 2026 [17]. The authors regard the fact-checking department of that newspaper as the noiselessness assumption.

4.5 Carathéodory and the dimension of Jensen

Theorem 7 (Carathéodory [2])**.** *If $S \subseteq \mathbb{R}^n$, every point of $\mathscr{J}(S)$ is a convex combination of at most $n+1$ points of S.*

Corollary 8 (Dimension of Jensen)**.** *If $\mathbb{E}[H]$ cannot be written with fewer than k generators, then $n \geq k-1$. With $k = 2$ (Lori, Alice), Jensen is at least one-dimensional. Conversely, in the plane no love polygon with more than three vertices is ever necessary.*

Definition 7 (Love triangle)**.** A *love triangle* is the 2-simplex $\mathrm{conv}\{j, \ell, \alpha\}$ spanned by three affinely independent Huangs.

Remark 9. Jensen is a *vertex* of the love triangle, while his expectation $\mathbb{E}[H] = \lambda\ell + (1-\lambda)\alpha$ lies on the edge *opposite* him. The distance from a vertex to the opposite edge is the height of the triangle; we call it the *moral height*, and note that it is maximised when Jensen is as far from both Huangs as possible, i.e. at work.

4.6 Responsibility is a vertex

Theorem 9 (Vertex in responsibility, interior in practice)**.** *Let r be a probability measure on $\{\mathit{Jensen}, \mathit{family}, \mathit{Board}\}$ describing the allocation of responsibility. Then $r = \delta_{\mathit{Jensen}}$, an extreme point of the simplex, whereas the Huang measure μ_λ with $\lambda \in (0,1)$ lies in the relative interior of its simplex.*

Proof. "I take full responsibility for what I did" [15] gives $r(\text{Jensen}) = 1$; the family "bears no responsibility" [15] gives $r(\text{family}) = 0$; the Board has "confidence in Jensen and his leadership team" [16], which we read as $r(\text{Board}) = 0$. The measure is a Dirac mass, hence a vertex. The second statement is the definition of the relative interior. □

Corollary 10. *Since r is a Dirac mass it is almost surely constant, so by Theorem 3 Jensen's inequality holds* with equality *for Jensen's responsibility. Responsibility and the jacket are the only two random variables in this paper with this property. Jensen is a vertex in responsibility and an interior point in practice; this is the first occasion on which he is known to have chosen an extreme point.*

4.7 The leather jacket stays on

Theorem 11 (The Leather Jacket Stays On)**.** *The Leather Jacket $\mathscr{J}$ is a closure operator on the subsets of $\mathbb{R}^d$:*

(i) $S \subseteq \mathscr{J}(S)$ (extensive: the jacket covers the Huangs);

(ii) $S \subseteq T \Rightarrow \mathscr{J}(S) \subseteq \mathscr{J}(T)$ (monotone: a bigger family needs a bigger jacket);

(iii) $\mathscr{J}(\mathscr{J}(S)) = \mathscr{J}(S)$ (idempotent: the jacket of the jacket is the jacket).

Moreover $\mathscr{J}(S) = S$ if and only if S is HUANG*, and if S is compact then $\mathscr{J}(S)$ is compact: the jacket zips up.*

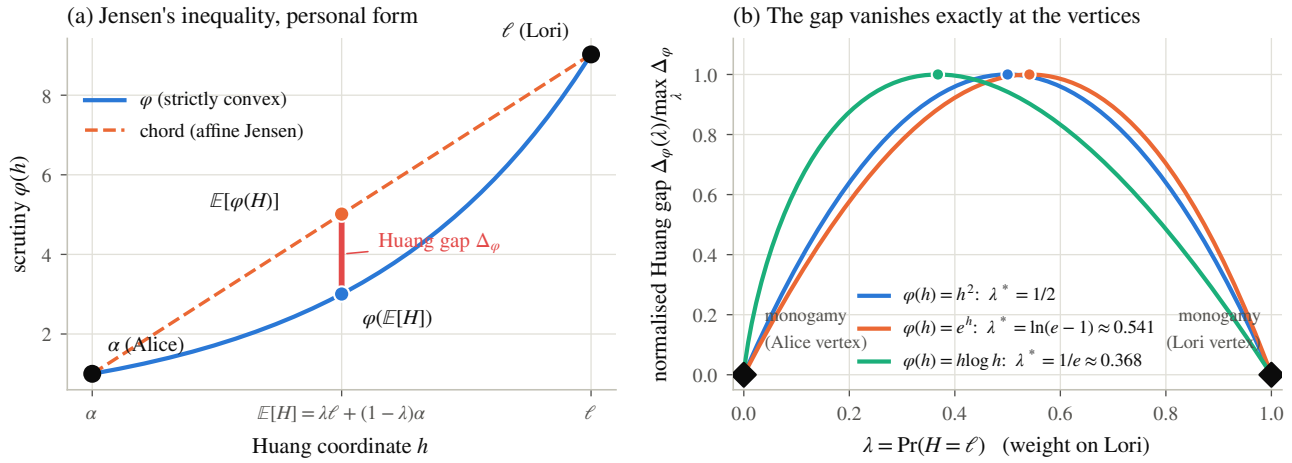


Figure 1: The Huang gap. (a) Jensen's inequality in personal form: for strictly convex scrutiny φ, the value at the expected Huang lies strictly below the expected scrutiny, and the vertical difference is Δ_φ. (b) The two-point Huang gap $\Delta_\varphi(\lambda)$, normalised by its maximum, for $\varphi(h) = h^2$ ($\Delta = \lambda(1-\lambda)$), $\varphi(h) = e^h$ ($\Delta = \lambda e + (1-\lambda) - e^\lambda$) and $\varphi(h) = h\log h$ ($\Delta = -\lambda\log\lambda$), with Alice at $h = 0$ and Lori at $h = 1$. All three vanish exactly at the vertices (monogamy) and are positive in the interior (Theorem 5); the maximiser λ^* depends on the scrutiny.

Algorithm 1 JENSEN: projected descent on the weight placed on Lori

Require: Huangs $\ell, \alpha \in \mathscr{H}$; step size η (Megaspeed); scrutiny φ
Ensure: a Huang $H \in \mathscr{J}(\{\ell, \alpha\}) \subseteq \mathscr{H}$ and a responsibility measure r

1: $\lambda \leftarrow 1$ ▷ 1985
2: **while** the Board has no reason to believe **do**
3: $\quad \lambda \leftarrow \mathrm{clip}\big(\lambda - \eta\, \partial_\lambda \varphi(\lambda\ell + (1-\lambda)\alpha), 0, 1\big)$ ▷ stay in the jacket
4: $\quad H \leftarrow \lambda\ell + (1-\lambda)\alpha$ ▷ the JENSEN step
5: $\quad$ fly to Taipei
6: **end while**
7: $r \leftarrow \delta_{\text{Jensen}}$ ▷ full responsibility (Theorem 9)
8: **return** H, r, the leather jacket

Proof. (i) and (ii) are immediate from Definition 5. (iii): $\mathscr{J}(S)$ is convex, and the convex hull of a convex set is itself. The characterisation of HUANG sets is the definition. Compactness: by Theorem 7, $\mathscr{J}(S)$ is the image of the compact set $\Delta_d \times S^{d+1}$ under the continuous map $(\lambda, x_0, \ldots, x_d) \mapsto \sum_i \lambda_i x_i$, hence compact. Once the jacket is on, applying the jacket again changes nothing: the leather jacket stays on. □

Remark 10 (Compactness of affairs). A set in $\mathbb{R}^d$ is compact if and only if it is closed and bounded. The affair was bounded (2024–2025) but is not closed (the investigation is ongoing). It is therefore not compact and, by Theorem 11, not a jacket.

Algorithm 1 summarises the procedure. The projection in line 3 is onto the 1-simplex, so every iterate is a convex combination and, by Theorem 1, a Huang; the loop condition is evaluated by the Board. The algorithm terminated on October 29, 2026.

5 Convexity Analysis with Chinese Characteristics

Jensen meets thousands of customers.
—NVIDIA, 2026 [19]

NVIDIA's defence (Hypothesis 4) is that the meetings in Taipei were "not unusual" [18] because Jensen meets thousands of customers. We model this defence with the appropriate stochastic process.

5.1 The Chinese Restaurant Process

In the Chinese Restaurant Process [6, 7] with concentration parameter $\alpha > 0$, customers enter a restaurant one at a time. Customer 1 sits at table 1. Customer $n+1$ joins an existing table k with probability $n_k/(n+\alpha)$, where n_k is the number of customers already seated there, and opens a *new* table with probability $\alpha/(n+\alpha)$. The process is exchangeable, exhibits a rich-get-richer dynamic, and underlies the Dirichlet process. We identify the concentration parameter with ALICE, for reasons of notation and concentration.

Let K_n be the number of tables after n customers. Since the indicator that customer i opens a new table is Bernoulli with parameter $\alpha/(\alpha+i-1)$,

$$\mathbb{E}[K_n] = \sum_{i=1}^{n} \frac{\alpha}{\alpha+i-1} = \alpha\log n + O(1). \tag{2}$$

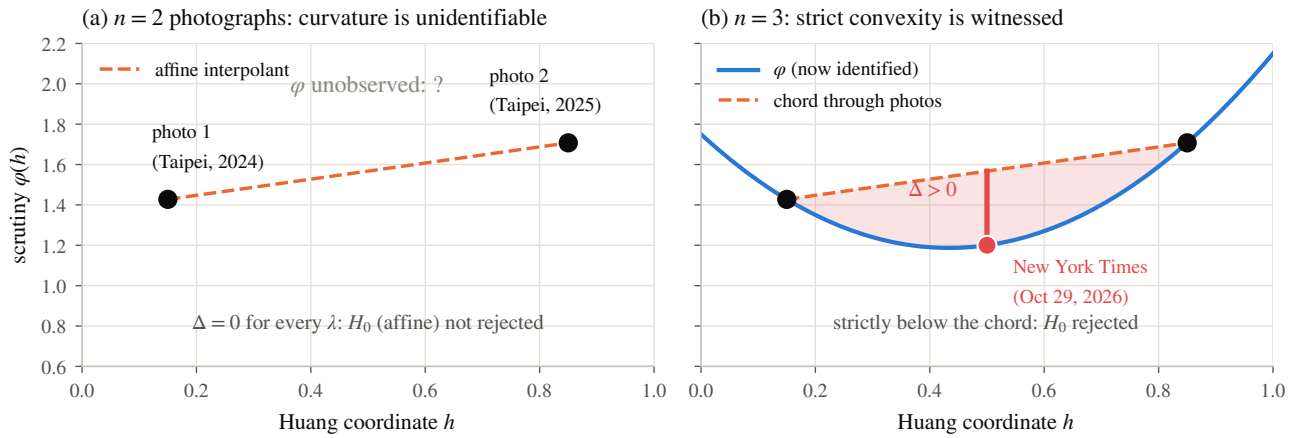


Figure 2: Curvature is undetectable from two photographs. (a) Two noiseless photographs are interpolated exactly by an affine function, under which the Huang gap is identically zero; Hypothesis 1 survives. (b) A third photograph lying strictly below the chord witnesses strict convexity and a positive gap (shaded); Hypothesis 1 is rejected with $p = 0$ (Theorem 6). The third point was supplied by the New York Times [17].

Proposition 12 (Thousands of customers)**.** *Under Hypothesis 4 with* $n = 5{,}000$*, the expected number of tables is* 1.9*,* 5.2*,* 9.1 *and* 16.2 *for* $\alpha = 0.1, 0.5, 1, 2$ *respectively (Figure 5a). In particular, the more customers Jensen meets, the more tables are opened in expectation, and the defence is monotonically self-defeating in n.*

Proof. Evaluate (2). Monotonicity is because every summand is positive. □

Remark 11. NVIDIA was founded in 1993 at a Denny's in East San Jose [22]. It is fitting that its chief executive's customer relations are governed by a restaurant process. We leave the estimation of α to the independent committee, noting only that the maximum-likelihood estimate given one observed new table is strictly positive.

Remark 12 (Rich get richer). Table 1 (est. 1985) seats the overwhelming majority of customers in every realisation (Figure 5b); the process is, in this sense, loyal. Nevertheless the probability $\alpha/(n+\alpha)$ of opening a new table is strictly positive for every n, and in Taipei it was realised.

5.2 The union bound

The defence can also be read as a union bound. If each of n customers independently presents an opportunity with probability p, then $\mathbb{P}(\text{at least one}) \le np$. For n in the thousands this bound exceeds 1 as soon as $p \ge 10^{-3}$, at which point it is vacuous. "Thousands of customers" is therefore a bound that proves nothing precisely because it is large.

5.3 Other Chinese results

The analysis above is not to be confused with the Chinese Remainder Theorem, which concerns what remains after Jensen returns from Taipei, nor with the Chinese Postman Problem [8], which asks for the shortest closed walk traversing every edge of a graph. For the love triangle (Definition 7) every vertex has degree 2, so the graph is Eulerian and the optimal postman tour is the love triangle itself, of length 3.

6 Experiments: Trivial by Jensen's

Our company is thirty days from going out of business.
—J. H. Huang, every year [22]

Setup. Our dataset consists of $n = 0$ flight logs, 2 photographs (Taipei, 2024 and 2025 [21]) and 1 newspaper [17]. All experiments were run on a single H100, which we define as a Huang at 100% fidelity. Code is available upon request from the independent committee.

The Huang gap. Figure 1 confirms Theorem 5: under every strictly convex scrutiny the gap is zero at both vertices and positive in between. The maximiser moves with the scrutiny—$\lambda^* = \frac{1}{2}$ for quadratic scrutiny, $\lambda^* = \ln(e-1) \approx 0.541$ for exponential scrutiny, and $\lambda^* = 1/e \approx 0.368$ for entropic scrutiny—but the vertices are the vertices regardless of φ. The fact that the maximiser is interior for *every* strictly convex φ is the mathematical content of the observation that there is no scrutiny under which an interior point looks like a vertex.

Three photographs. Figure 2 realises Theorem 6. We emphasise that panel (a) is not a failure of the photographer: no finite amount of staring at two points can reveal curvature,

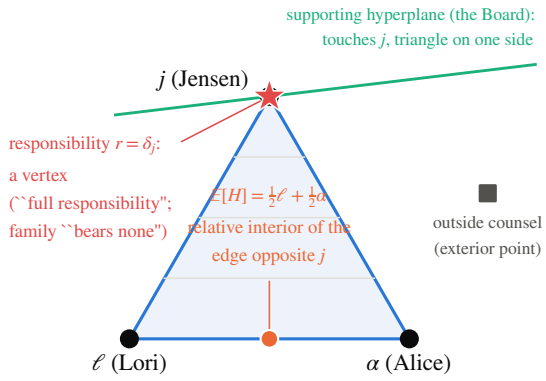


Figure 3: The love triangle $\mathrm{conv}\{j, \ell, \alpha\}$, a 2-simplex. Responsibility is the Dirac mass at the vertex j (Theorem 9), while the mean Huang lies in the relative interior of the opposite edge. The Board is a supporting hyperplane: it touches Jensen at a single point and the whole triangle lies on one side of it. Outside counsel is an exterior point and is therefore strictly separable from the triangle. By Carathéodory (Theorem 7), three generators suffice in the plane.

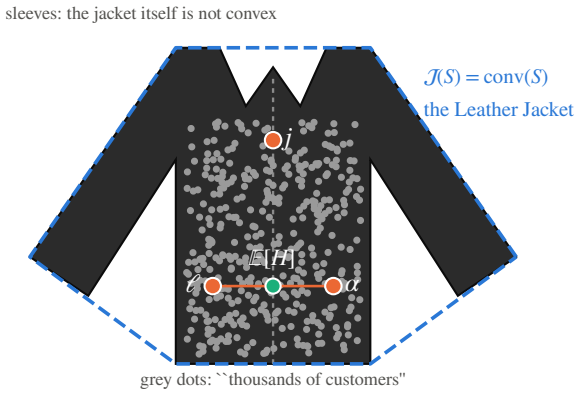


Figure 4: The Leather Jacket. The jacket itself (black) is not convex: it has sleeves. Its convex hull $\mathcal{J}(S)$ (dashed) is the smallest convex set containing it, computed by Quickhull. The hull contains ℓ, α, j, the mean Huang $\mathbb{E}[H]$, and thousands of customers (grey). Applying $\mathcal{J}$ again returns the same dashed polygon (Theorem 11): the leather jacket stays on.

which is why the photographs were "not unusual" for two years. Curvature is a second-order quantity and requires a third point.

The love triangle. Figure 3 draws the 2-simplex. The supporting-hyperplane theorem [3] guarantees that every boundary point of a convex set admits a hyperplane through it with the set on one side; the Board's statement of confidence [16] is exactly such a hyperplane through the vertex j. Outside counsel, being an exterior point, can be *strictly* separated from the closed convex triangle, which is the entire point of outside counsel.

The Leather Jacket. Figure 4 computes $\mathcal{J}$ of a jacket. Two features deserve comment. First, the jacket is not itself convex—sleeves are a failure of convexity, since the segment joining the two cuffs leaves the garment—so the jacket is strictly contained in its hull. Second, the hull is idempotent: we recomputed $\mathcal{J}(\mathcal{J}(S))$ and obtained the same polygon to machine precision, which is the strongest numerical evidence we are able to offer that the leather jacket stays on.

The restaurant. Figure 5 evaluates (2) and simulates one realisation. The realisation was selected to have K_n equal to the rounded expectation, which the authors consider a form of regularisation.

Tables. Table 1 compares Jensen's inequality with related inequalities and with Jensen. Table 2 compares Huangs. We draw the reader's attention to the "Stays on" row.

7 Discussion

The more you buy, the more you save.
—J. H. Huang, GTC keynotes

7.1 The Board

The Board "has confidence in Jensen" [16]. A confidence interval of the form $[\text{Jensen}, \text{Jensen}]$ has width zero, so the Board's belief about Jensen is almost surely constant, so by Theorem 3 Jensen's inequality holds for it with equality, which contradicts Corollary 4. The Board's confidence and Jensen's memo cannot both be sharp. We leave the resolution to the committee, which is *independent*, i.e. for every event A concerning what Jensen did and every event B concerning what the committee finds, $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. Moreover, writing $\mathscr{F}_{\text{pub}} \subseteq \mathscr{F}_{\text{Board}}$ for the public and Board σ-algebras, the tower property gives

$$\mathbb{E}\big[\mathbb{E}[X \mid \mathscr{F}_{\text{Board}}] \,\big|\, \mathscr{F}_{\text{pub}}\big] = \mathbb{E}[X \mid \mathscr{F}_{\text{pub}}] :$$

the expected outcome of the investigation, from the public's point of view, is exactly what the public already expects. The

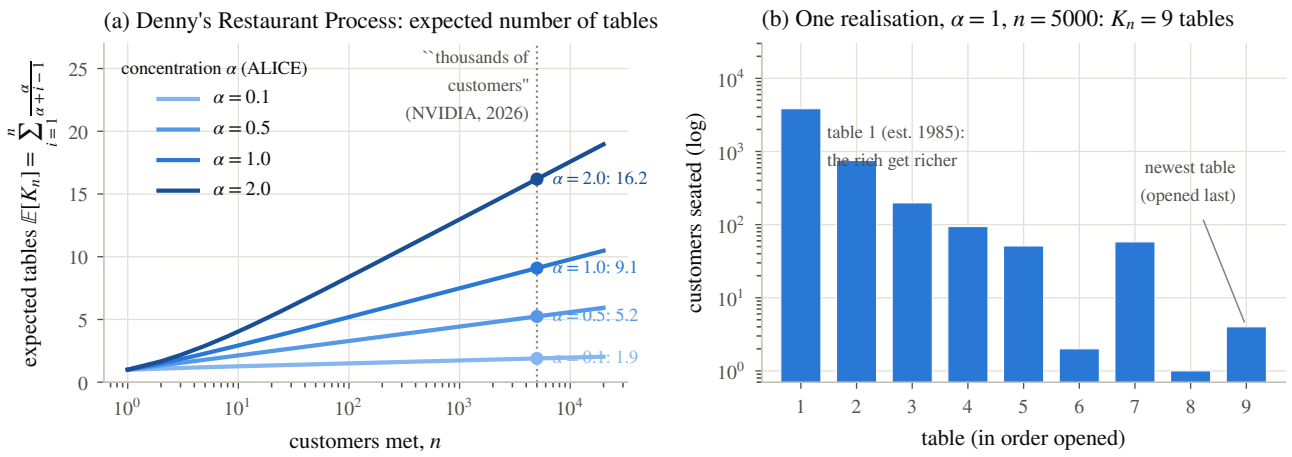


Figure 5: Convexity analysis with Chinese characteristics. (a) Expected number of tables $\mathbb{E}[K_n]$ in the Chinese Restaurant Process, Eq. (2), for four concentration parameters; at $n = 5{,}000$ ("thousands of customers") the expectation is 1.9, 5.2, 9.1 and 16.2. (b) One realisation with $\alpha = 1$ and $n = 5{,}000$ customers: $K_n = 9$ tables, with table 1 seating the large majority (rich get richer) and the newest table seating four.

Table 1: Comparison of inequalities. Jensen Huang is the only entry to be investigated by Congress, and the only one with a conjugate he did not use.

	Jensen's	**Hölder's**	**Young's**	**Markov's**	**Huang's Law**	**Jensen Huang**
Named after an engineer	✓	✗	✗	✗	✓	✓
Is an inequality	✓	✓	✓	✓	✗	✓
Strict unless a.s. constant	✓	✗	✗	✗	✗	✓
Tight	✓	✓	✓	✓	✗	✓
Involves a conjugate	✗	✓	✓	✗	✗	✓
Holds in expectation	✓	✗	✗	✓	✓	✗
Admitted	✗	✗	✗	✗	✗	✓
Investigated by Congress	✗	✗	✗	✗	✗	✓
Trivial by Jensen's	✓	✓	✓	✗	✗	✓
Hotel?	Trivago	Trivago	Trivago	Trivago	Trivago	Trivago

Table 2: Comparison of Huangs. K denotes the leather jacket.

	ℓ (Lori)	α (Alice)	j (Jensen)	K
Is a Huang	✓	✓	✓	✗[†]
Convex	✓	✓	✗	✗[‡]
Almost surely constant	✓	?	✗	✓
Whereabouts known	✓	✗	✓	✓[§]
Lab partner	✓	✗	✗	✗
Global minimum	✓	✗	✗	✗
Interior point	✗	✓	✓	✗
Bears responsibility	0	0	1	0
Stays on	✓	✗	✗	✓

[†]It is a jacket. [‡]Sleeves. [§]On Jensen.

Board has also undertaken to "examine all available information" [16]; conditional Jensen, $\varphi(\mathbb{E}[H \mid \mathscr{G}]) \le \mathbb{E}[\varphi(H) \mid \mathscr{G}]$ almost surely, holds for every sub-σ-algebra $\mathscr{G}$, so the inequality survives whatever the Board conditions on.

7.2 Exposed faces

A face F of a convex body C is *exposed* if $F = C \cap P$ for some supporting hyperplane P [3]. Jensen's face was exposed on October 29, 2026, by a hyperplane supported by the New York Times [17].

7.3 The more you buy, the more you save

Claim 13. *If the cost of n GPUs is $C(n)$ with C concave and $C(0) = 0$, then the average cost $C(n)/n$ is non-increasing in n: the more you buy, the more you save.*

Proof. For $0 < m < n$, write $m = \frac{m}{n}\,n + \left(1 - \frac{m}{n}\right)0$. By concavity, $C(m) \ge \frac{m}{n}C(n) + \left(1 - \frac{m}{n}\right)C(0) = \frac{m}{n}C(n)$, i.e. $C(m)/m \ge C(n)/n$. □

This is Jensen's inequality (in its concave form) applied to a two-point distribution on $\{0, n\}$ with weight m/n on n. It is the only inequality in this paper that Jensen has consistently respected, and we note that it, too, is a statement about convex combinations.

7.4 Threats to validity

(1) We model Jensen as a random variable. He may instead be a stochastic process, in which case the relevant object is a filtration $(\mathscr{F}_t)_{t\ge 0}$, a briefing on which a group of House committee chairs has requested, with a November 11 deadline to propose a date [21].

(2) We assume scrutiny is strictly convex. If φ is affine, Jensen is faithful and this paper is vacuous. We regard this as unlikely [15].

(3) The marriage is 41 years old. Since 41 is prime, the marriage admits no nontrivial factorisation; nevertheless a nontrivial divisor was found in Taipei. This contradicts the fundamental theorem of arithmetic and is left to future work.

(4) Our photographs are assumed noiseless. If they are noisy, Theorem 6(b) holds with a p-value that is small rather than zero, which is the usual situation in science and, we are told, in marriage.

8 Future Work

Four sequels are in preparation. *ELBO: The Evidence Lower BOard* applies variational inference to the Board's statement, which is a lower bound on the evidence whose gap is exactly the KL divergence between the official narrative and the true posterior. *Honeypots Are Non-Convex* observes that a spurious local minimum can exist only in a non-convex landscape. *Conjugates* studies the Fenchel conjugate f^{*} [5]: since $f^{**}=f$ only for proper closed convex f, and Jensen is improper, the conjugate of Lori's conjugate is not Jensen but his convex envelope—the best convex man he could have been. *Mixed-Fidelity Training* notes that half precision carries only ten mantissa bits of fidelity.

9 Conclusion

We have shown that the Huangs form a convex set, that Jensen's inequality is strict for every Huang that is not almost surely constant, that two photographs cannot detect curvature but three can, that responsibility is a vertex and practice an interior point, that a love triangle is a 2-simplex, that thousands of customers open $\alpha \log n$ tables in expectation, and that the hull operator is idempotent.

The leather jacket stays on.

Ethics Statement

All quotations attributed to real people are real and cited. All allegations are alleged; nobody in this paper has been charged with anything, including the authors. Lori is the global minimum. No Huangs were harmed in the writing of this paper; one was allegedly compromised, which the Board has no reason to believe.

Acknowledgements

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A Proofs

Proof of Theorem 3. Let $m=\mathbb{E}[H]$, which lies in the convex domain of φ by Theorem 1. Since φ is convex it has a subgradient $g\in\partial\varphi(m)$, so that $\varphi(h)\ge\varphi(m)+\langle g,h-m\rangle$ for all h. Substituting $h=H$ and taking expectations, the linear term vanishes and $\mathbb{E}[\varphi(H)]\ge\varphi(m)$. If equality holds, then $\varphi(H)=\varphi(m)+\langle g,H-m\rangle$ almost surely, i.e. φ agrees with its supporting affine function at H; for strictly convex φ the supporting function touches φ only at m, so $H=m$ almost surely. Conversely, a constant H trivially gives equality. $\square$

Proof of Theorem 5. (i) At $\lambda\in\{0,1\}$ the measure is a point mass and both sides coincide. (ii) The first two terms of Δ_φ are affine in λ and the last is the negative of a convex function of an affine function of λ, hence concave; so Δ_φ is concave. It is nonnegative by convexity of φ and, for $\ell\neq\alpha$ and $\lambda\in(0,1)$, strictly positive by strict convexity. A concave function vanishing at both endpoints and positive inside attains its maximum in the interior. (iii) With $\mu=\mathbb{E}[H]$, $\mathbb{E}[D_\varphi(H,\mu)]=\mathbb{E}[\varphi(H)]-\varphi(\mu)-\langle\nabla\varphi(\mu),\mathbb{E}[H]-\mu\rangle=\mathbb{E}[\varphi(H)]-\varphi(\mu)$. (iv) For $\varphi=\|\cdot\|^2$ the Bregman divergence is $\|x-y\|^2$, so $\Delta_\varphi=\mathbb{E}\|H-\mu\|^2=\mathrm{Var}(H)$; for a two-point measure this is $\lambda\|\ell-\mu\|^2+(1-\lambda)\|\alpha-\mu\|^2$ with $\ell-\mu=(1-\lambda)(\ell-\alpha)$ and $\alpha-\mu=-\lambda(\ell-\alpha)$, giving $\lambda(1-\lambda)\|\ell-\alpha\|^2$. The map $\lambda\mapsto\lambda(1-\lambda)$ is maximised at $\tfrac12$ with value $\tfrac14$. (v) By Taylor's theorem with integral remainder, $D_\varphi(h,\mu)=\int_0^1(1-t)\varphi''(\mu+t(h-\mu))\,dt\,(h-\mu)^2$, which lies between $\tfrac{m}{2}(h-\mu)^2$ and $\tfrac{M}{2}(h-\mu)^2$; take expectations and apply (iii). $\square$

Proof of Theorem 11, compactness. $\Delta_d=\{\lambda\in\mathbb{R}^{d+1}:\lambda_i\ge 0,\sum_i\lambda_i=1\}$ is compact, and S^{d+1} is compact by Tychonoff (finitely many factors suffice). By Carathéodory, $\mathscr{J}(S)$ equals the image of $\Delta_d\times S^{d+1}$ under the continuous map $(\lambda,x)\mapsto\sum_i\lambda_i x_i$. Continuous images of compact sets are compact. The jacket zips up. $\square$

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